



Chapter 5 – SIEM (Security Information and Event Management) & Splunk



Introduction to SIEM

What is SIEM?

SIEM stands for:



Security Information and Event Management

SIEM is one of the most important technologies used inside a **Security Operations Center (SOC)**.

Its main purpose is to:

- ✓ Collect logs from different devices
 - ✓ Analyze security events
 - ✓ Detect suspicious activities
 - ✓ Generate alerts
 - ✓ Help SOC Analysts investigate attacks
 - ✓ Provide centralized visibility of the entire IT infrastructure
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Why SIEM is Important in a SOC?

A company may have:



Routers



Switches



Firewalls



IDS (Intrusion Detection Systems)



IPS (Intrusion Prevention Systems)



Windows Systems

-  Mac Systems
-  Mobile Devices
-  Printers
-  CCTV Systems
-  Windows Servers
-  Linux Servers
-  AWS Cloud
-  Azure Cloud
-  Office 365
-  Active Directory

Monitoring all these devices manually is impossible.

Therefore, SIEM collects logs from all devices and displays them in one centralized location.

Real-Life Example

Imagine you are the security guard of a shopping mall.

The mall contains:

-  100 Shops
-  500 Cameras
-  Hundreds of Doors
-  Thousands of Visitors

You cannot physically monitor everything.

Instead, all CCTV feeds are sent to one monitoring room.

SIEM works exactly the same way.

It acts as the central monitoring room of an organization's IT infrastructure.

What Does a SOC Analyst Do in a SIEM?

The primary job of a SOC Analyst is monitoring.

The analyst continuously looks for:

-  Failed Login Attempts
 -  Malware Activity
 -  Suspicious Network Traffic
 -  Unauthorized Access
 -  Privilege Escalation
 -  Brute Force Attacks
 -  Data Exfiltration
 -  Policy Violations
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Understanding Events and Incidents

Event

An event is any activity recorded in logs.

Examples:

-  User Login
 -  File Access
 -  Service Started
 -  Website Opened
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Security Incident

A security incident is a suspicious or malicious event that requires investigation.

Examples:

 Multiple Failed Login Attempts

 Malware Execution

 Unauthorized Access

 Data Theft

Example

Normal Event:

User Login Successful

Security Incident:

50 Failed Password Attempts Within 2 Minutes

This may indicate a brute-force attack.

Rule-Based Monitoring in SIEM

SOC Analysts create detection rules.

These rules automatically generate alerts.

Example 1 – Failed Login Rule

Rule:

If Failed Login Attempts > 5

Generate Alert

Example 2 – Traffic Spike Rule

Rule:

If More Than 1000 Packets Arrive Within 5 Seconds

Generate Alert

This may indicate:

- ⚠️ DDoS Attack
 - ⚠️ Port Scan
 - ⚠️ Bot Activity
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Example 3 – Threat Intelligence Rule

Suppose Threat Intelligence provides:

- 🌐 Malicious IP
- 📁 Malicious Hash
- 🌐 Malicious Domain

The SOC Analyst creates a SIEM rule:

If this IP appears in our environment,
Generate Alert

or

If this malware hash is detected,
Generate Alert

Relationship Between Threat Intelligence and SIEM

Threat Intelligence Team

Responsibilities:

- 🔍 Find malicious IPs
 - 🔍 Find malware hashes
 - 🔍 Find malicious domains
 - 🔍 Research attacker infrastructure
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SOC Team

Responsibilities:

- 🛡️ Create SIEM Detection Rules
 - 🛡️ Generate Alerts
 - 🛡️ Monitor Security Events
 - 🛡️ Investigate Incidents
-

Example Workflow

Threat Researcher discovers:

IP: 185.XX.XX.XX

associated with ransomware.

SOC Team creates:

Alert if IP 185.XX.XX.XX appears anywhere in logs.

This is a common real-world SOC workflow.



What is Splunk?

Definition

Splunk is a powerful platform used to collect, analyze, search, visualize, and monitor machine-generated data (logs) in real time.

Originally Splunk was a log analysis platform.

Today it is widely used as a SIEM solution.

[Splunk Official Website](https://splunk.com)

Why Splunk is Used?

Splunk helps organizations:

- 🔍 Monitor Logs

 Detect Security Threats

 Create Dashboards

 Generate Reports

 Visualize Data

 Correlate Events

 Investigate Incidents

Why Manual Log Analysis is Difficult

Imagine a company with:

 500 Computers

 50 Servers

 20 Firewalls

 100 Network Devices

Every day these devices generate millions of logs.

Manually checking logs is practically impossible.

Splunk solves this problem.

What Are Logs?

Logs are automatically generated records of activities occurring inside systems.

They help security teams understand:

Who did what?

When did it happen?

Where did it happen?

Was it successful?

Was it malicious?

Types of Logs in Splunk

1 System Logs

Contain operating system activities.

Examples:

- User Logins
 - Software Installation
 - Windows Updates
 - Browser Activities
-

2 Application Logs

Generated by applications.

Examples:

- Microsoft Office
 - Apache
 - IIS
 - Antivirus Software
-

3 Security Logs

Contain security-related events.

Examples:

- Authentication Logs
- Firewall Logs
- IDS Alerts
- IPS Alerts

Network Logs

Generated by:

-  Routers
-  Switches
-  Firewalls
-  VPN Devices
-  Proxy Servers

Database Logs

Contain:

- SQL Queries
- Database Errors
- Transactions

Server Logs

Contain:

- CPU Usage
- Memory Usage
- Service Status
- Access Requests

Active Directory Logs

Contain:

-  Login Events

 Password Changes

 Privilege Changes

 User Creation

Splunk Architecture

Splunk consists of three major components.

1. Forwarder

What is a Forwarder?

A Forwarder is a lightweight application installed on source devices.

Its job is to collect logs and send them to Splunk.

Example

Windows Machine



Splunk Universal Forwarder



Splunk Server

Purpose

- ☒ Collect Logs
 - ☒ Forward Logs
 - ☒ Real-Time Data Transfer
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2. Indexer

What is an Indexer?

The Indexer receives logs from Forwarders.

Its job is to:

-  Receive Logs
-  Organize Logs
-  Index Logs
-  Store Logs

Example

Suppose logs arrive from:

- 10 Windows Servers
- 5 Linux Servers
- 3 Firewalls

Indexer organizes all data into searchable format.

3. Search Head

What is Search Head?

Search Head is the user interface used by analysts.

It provides:

-  Search Capability
-  Dashboards
-  Alerts
-  Reports
-  Visualizations

Data Flow in Splunk

Endpoint
↓
Forwarder
↓
Indexer
↓
Search Head
↓
SOC Analyst

Source-Based Indexing

Splunk separates logs into different indexes.

Examples:

index=windows_security

index=firewall_logs

index=linux_logs

index=webserver_logs

Example Query

index=webserver_logs error OR warning

This displays all errors and warnings from web server logs.

Searching and Reporting

One of the most important tabs in Splunk is:

Searching & Reporting

Here analysts write SPL queries.

SPL stands for:

Search Processing Language

It works somewhat similar to SQL.

Example Tasks

Find:

- Failed Logins
 - Malware Alerts
 - Firewall Events
 - Router Logs
 - VPN Activity
-

Time-Based Searches

Analysts can search logs based on:

-  Last 15 Minutes
 -  Last Hour
 -  Last 24 Hours
 -  Last 7 Days
 -  Custom Time Range
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Dashboards in Splunk

What is a Dashboard?

A dashboard visually displays important security information.

Examples:

-  Login Activity
-  Malware Detections

 Firewall Blocks

 VPN Connections

 Security Incidents

Example Dashboard

SOC Dashboard may display:

 Top 10 Attacking IPs

 Failed Logins

 Malware Detections

 Active Incidents

Alerts in Splunk

What is an Alert?

An alert is automatically generated when a rule condition is met.

Example

Failed Login Attempts > 5



 Alert Generated

Example

Malicious Hash Detected



 Alert Generated

SOC Investigation Process

Suppose Splunk generates an alert.

Step 1

Identify:

-  Source IP
-  Destination IP
-  Username
-  Hostname

Step 2

Access the affected machine.

Step 3

Check:

-  Running Files
-  Running Processes
-  DLL Files
-  Scheduled Tasks
-  Registry Entries

Step 4

Calculate File Hash.

Step 5

Check reputation using: [VirusTotal](https://www.virustotal.com/)

Step 6

Determine:

Was it malware?

Was it a false positive?

Was there actual compromise?

SOC Analyst Levels

L1 SOC Analyst

Responsibilities:

- 👁️ Monitoring
 - 👁️ Alert Review
 - 👁️ Ticket Creation
 - 👁️ Basic Investigation
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L2 SOC Analyst

Responsibilities:

- 🔍 Deep Investigation
 - 🔍 Malware Analysis
 - 🔍 Incident Validation
 - 🔍 Threat Hunting
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L3 SOC Analyst

Responsibilities:

-  Advanced Threat Hunting
 -  Incident Response
 -  Threat Research
 -  Detection Engineering
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Splunk Installation Overview

Splunk Enterprise Installation

Download from:

[Splunk Enterprise Downloads](#)

Windows Installation

- 1 Run MSI Installer
- 2 Choose Installation Path

C:\Program Files\Splunk

- 3 Create Admin Account
- 4 Access GUI

<http://localhost:8000>

Linux Installation

Extract package and start Splunk service.

Commonly installed on Linux servers in enterprise environments.

Installing Splunk Universal Forwarder

Install on:

 Windows Systems

 Linux Systems

 Servers

Then configure:

Forwarder → Splunk Indexer

Default receiving port:

9997

Integration with Cloud Platforms

Splunk can integrate with:

 Microsoft 365

 Azure

 Intune

 AWS

 Google Cloud

Benefits of Integration

Monitor:

 Email Security

 Login Activity

 Device Compliance

 Cloud Access

 Azure AD Events

Interview Questions

Q1. What is SIEM?

SIEM (Security Information and Event Management) is a platform that collects, analyzes, correlates, and monitors security events from multiple devices to detect threats and generate alerts.

Q2. What are the three major Splunk components?

- ✓ Forwarder
 - ✓ Indexer
 - ✓ Search Head
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Q3. What is a Forwarder?

A Forwarder collects logs from source systems and sends them to the Splunk Indexer.

Q4. What is an Indexer?

An Indexer receives, stores, and organizes logs into searchable data.

Q5. What is SPL?

SPL (Search Processing Language) is Splunk's query language used for searching and analyzing log data.

Q6. What is the difference between an Event and an Incident?

Event	Incident
Normal activity	Suspicious activity
Login	Brute-force attack

Event	Incident
File access	Malware execution

Chapter 5 Summary

SIEM (Security Information and Event Management) is the heart of SOC operations. It centralizes logs from routers, switches, firewalls, IDS/IPS, servers, endpoints, cloud platforms, and applications. Splunk is one of the most widely used SIEM platforms that collects logs using Forwarders, stores them through Indexers, and allows analysts to investigate data through the Search Head. SOC Analysts create detection rules, generate alerts, build dashboards, investigate incidents, and use Threat Intelligence data to identify malicious activity. Understanding Splunk architecture, alerts, dashboards, log analysis, and incident investigation is one of the most important skills required for SOC Analyst interviews and real-world security operations.       